SEAA 4243 FINITE ELEMENT METHOD · 2L + 1T

# Plane Stress Part 2 of 3

[B] = [∂]ᵀ[N], the 8 × 8 stiffness matrix [k] = t∫[B]ᵀ[E][B] dA, nodal loads, assembly with two dofs per node, and the cantilever solved by hand

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Text: Mohd Yassin et al., *Two-Dimensional Finite Element Formulations for Heat, Solids and Fluids (with Matlab)*, MSNM 2020 — §3.2.2 – §3.4

ACTIVITY 0 · INDIVIDUAL · RETRIEVAL WARM-UP

# Three questions from the last session — on paper

5 MIN

**Write down**

- The plane-stress PDE in matrix form and the sizes of $[\partial]$ and [E].
- The three strain components in terms of u and v.
- The dof numbers of global node 5 in the 2 × 2 cantilever mesh, and the dof list of element 1.

**WHY** Question 3 is the bookkeeping that today's assembly runs on.

5 minutes. Then the outcomes.

LEARNING OUTCOMES · TODAY

# By the end of this session you should be able to

1. Apply Galerkin and integration by parts to the plane-stress PDE and identify [k], {q} and {b}; write $[B] = [\partial]^T[N]$ and $[k] = t\int[B]^T[E][B]\,dA$.
2. Evaluate the 8 × 8 stiffness matrix of a four-node rectangular element from the closed-form entries $s_1 \ldots s_{12}$.
3. Convert an edge traction into equivalent nodal loads, assemble the global system with two dofs per node and impose displacement conditions.
4. Solve the 2 × 2 cantilever, compare with a refined mesh and with beam theory, and explain why the coarse mesh is too stiff.

**WHY** Text §3.2.2 – §3.4. Outcome 4 is the lesson the project depends on: a coarse Q4 mesh underestimates bending deflection badly.

SESSION MAP · 180 MINUTES

# Six blocks, seven short activities, one break

**0:00**
**Weak form and [B]**
30 min
A1 · 8 min

**0:30**
**The 8 × 8 [k]**
35 min
A2 · 8 min

**1:05**
**Load vectors**
20 min
A3 · 6 min

**1:25**
**Break**
10 min
no activity

**1:35**
**Assembly & supports**
40 min
A4 + A5

**2:15**
**Solution & checks**
25 min
A6 + QR

**2:40**
**MATLAB & wrap-up**
20 min
Exit ticket

## Bring forward

- Plane Stress Part 1 Activity 3: [E] = 224.44, 74.07, 75.19 (× $10^6$ N m$^{-2}$).
- Plane Stress Part 1 Activity 4: which dofs are fixed ($U_1$, $U_2$, $U_7$, $U_8$, $U_{13}$, $U_{14}$).
- The Heat Transfer Part 2 MATLAB script.

## The running example

- Cantilever 4 m × 2 m, t = 0.01 m, E = 200 × $10^6$ N m$^{-2}$, ν = 0.33, $q_y$ = 5 kN m$^{-1}$ downward on the free end.
- 2 × 2 mesh: a = 2 m, b = 1 m, 18 dofs.
- Text §3.4, equations (3.44)–(3.49), Table 3.1.

1

BLOCK 1 · 0:00 – 0:30

# Weak form and the matrix [B]

Substitute $\{u\}^T = [N]\{\hat{u}\}^T$, weight with $[N]^T$, integrate by parts. The strain–displacement matrix [B] appears twice in [k]. Text §3.2.2 – §3.3.

# Eight unknowns per element, eight weighted equations, one boundary term

$$[\partial][E][\partial]^T[N]\{\hat{u}\}^T \neq -\{f\} \quad (3.31)$$

$$\int_x\int_y [N]^T\Big([\partial][E][\partial]^T[N]\{\hat{u}\}^T + \{f\}\Big)\,dy\,dx = 0 \quad (3.32)$$

$$\underbrace{\int_x\int_y [N]^T[\partial][E][\partial]^T[N]\,dy\,dx}_{[k]\ (8\times 8)}\{\hat{u}\}^T = \underbrace{\int_x\int_y [N]^T\{f\}\,dy\,dx}_{\{q\}} + \underbrace{\int_s [N]^T\{b\}^T\,ds}_{\{b\}} \quad (3.34)–(3.38)$$

$$[k]\{\hat{u}\}^T = \{q\} + \{b\} = \{r\} \quad (3.38)$$

**WHY** $[N]^T$ is 8 × 2, so weighting gives 8 equations. Integration by parts moves one $[\partial]$ onto $[N]^T$ (left to right rule) and produces the traction boundary term $\{b\}$.

# [B] = [∂]ᵀ[N]: strains from the nodal displacements

$$\{\varepsilon\}^T = [\partial]^T[N]\{\hat{u}\}^T = [B]\{\hat{u}\}^T, \qquad \{\sigma\}^T = [E][B]\{\hat{u}\}^T$$

$$[B] = \begin{bmatrix} \frac{y-b}{ab} & 0 & \frac{b-y}{ab} & 0 & \frac{y}{ab} & 0 & \frac{-y}{ab} & 0 \\ 0 & \frac{x-a}{ab} & 0 & \frac{-x}{ab} & 0 & \frac{x}{ab} & 0 & \frac{a-x}{ab} \\ \frac{x-a}{ab} & \frac{y-b}{ab} & \frac{-x}{ab} & \frac{b-y}{ab} & \frac{x}{ab} & \frac{y}{ab} & \frac{a-x}{ab} & \frac{-y}{ab} \end{bmatrix}$$

**WHY** Row 1: $\partial N_i/\partial x$ under each $u_i$. Row 2: $\partial N_i/\partial y$ under each $v_i$. Row 3: $\partial N_i/\partial y$ under $u_i$, $\partial N_i/\partial x$ under $v_i$.

**CHECK** Each entry is a Heat Transfer Part 1 derivative: $\partial N_1/\partial x = -(b - y)/ab = (y - b)/ab$, and so on. Nothing new is computed.

# The same [B] twice, with [E] between them, integrated over the element and multiplied by the thickness

$$[k] = t \int_x \int_y [B]^T [E][B]\, dy\, dx \qquad (8 \times 3)(3 \times 3)(3 \times 8) = 8 \times 8$$

### Why the thickness appears

The integrals of (3.35) are per unit thickness; the physical element has thickness t, so every entry of [k] and every load is multiplied by t. The text keeps t inside $s_1 \ldots s_{12}$.

### Why it is symmetric

$[B]^T[E][B]$ is symmetric whenever [E] is. Integration preserves symmetry. The 8 × 8 matrix therefore has at most 36 distinct entries — and by the rectangle's symmetry only 12.

### What is being computed

$k_{ij}$ = force at dof i per unit displacement of dof j, all others held. It is the virtual-work pairing of the strain patterns of dofs i and j through Hooke's law.

ACTIVITY 1 · PAIRS · DERIVATION

# Build [B] at a point and check it

8 MIN

**Do**

- a = 2 m, b = 1 m, point (1.5, 0.3) — the Heat Transfer Part 1 point. Write $\partial N_i/\partial x$ and $\partial N_i/\partial y$ for i = 1…4 (you have them: −0.35, 0.35, 0.15, −0.15 and −0.25, −0.75, 0.75, 0.25).
- Fill the 3 × 8 matrix [B] with these numbers.
- Check: the sum of each row is zero. Why must it be?

$$[B] = \begin{bmatrix} \frac{y-b}{ab} & 0 & \frac{b-y}{ab} & 0 & \frac{y}{ab} & 0 & \frac{-y}{ab} & 0 \\ 0 & \frac{x-a}{ab} & 0 & \frac{-x}{ab} & 0 & \frac{x}{ab} & 0 & \frac{a-x}{ab} \\ \frac{x-a}{ab} & \frac{y-b}{ab} & \frac{-x}{ab} & \frac{b-y}{ab} & \frac{x}{ab} & \frac{y}{ab} & \frac{a-x}{ab} & \frac{-y}{ab} \end{bmatrix}$$

**CHECK** Row sums zero: a rigid translation (all $u_i$ equal, all $v_i$ equal) produces no strain.

8 minutes. Keep the sheet: Plane Stress Part 3 multiplies this [B] by the nodal displacements.

2

BLOCK 2 · 0:30 – 1:05

# The 8 × 8 stiffness matrix

$k_{34}$ in three steps, then the pattern of twelve closed-form values, then the numbers for the cantilever element. Text (3.39)–(3.44).

BLOCK 2 · $K_{34}$, STEP 1 · TEXT (3.39)

# Row 3 of $[N]^T[\partial]$ (dof $u_2$) times [E]

$$\cdots \begin{bmatrix} \vdots & \vdots & \vdots \\ \dfrac{\partial N_2}{\partial x} & 0 & \dfrac{\partial N_2}{\partial y} \\ \vdots & \vdots & \vdots \end{bmatrix} \frac{E}{1-\nu^2} \begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \dfrac{1-\nu}{2} \end{bmatrix} \cdots \tag{3.39}$$

$$\Rightarrow \quad \cdots \left[ \frac{E}{1-\nu^2}\frac{\partial N_2}{\partial x} \quad \frac{E\nu}{1-\nu^2}\frac{\partial N_2}{\partial x} \quad \frac{E(1-\nu)}{2(1-\nu^2)}\frac{\partial N_2}{\partial y} \right] \cdots$$

**WHY** Row 3 of the 8 × 3 matrix $[N]^T[\partial]$ belongs to $u_2$: the derivatives of $N_2$ sit in columns 1 and 3 ($\varepsilon_{xx}$ and $\varepsilon_{xy}$). Multiplying by [E] attaches a material constant to each.

# Times column 4 of $[\partial]^T[N]$ (dof $v_2$): two products survive

$$\cdots \left[ \frac{E}{1-\nu^2}\frac{\partial N_2}{\partial x} \quad \frac{E\nu}{1-\nu^2}\frac{\partial N_2}{\partial x} \quad \frac{E(1-\nu)}{2(1-\nu^2)}\frac{\partial N_2}{\partial y} \right] \begin{bmatrix} \cdots & 0 & \cdots \\ \cdots & \dfrac{\partial N_2}{\partial y} & \cdots \\ \cdots & \dfrac{\partial N_2}{\partial x} & \cdots \end{bmatrix} \tag{3.40}$$

$$\Rightarrow \quad \cdots \left[ \frac{E\nu}{1-\nu^2}\frac{\partial N_2}{\partial x}\frac{\partial N_2}{\partial y} + \frac{E(1-\nu)}{2(1-\nu^2)}\frac{\partial N_2}{\partial y}\frac{\partial N_2}{\partial x} \right] \cdots$$

**WHY** Column 4 belongs to $v_2$: $\partial N_2/\partial y$ in row 2 and $\partial N_2/\partial x$ in row 3, zero in row 1. The first product vanishes; the v-term and the shear term remain.

BLOCK 2 · $K_{34}$, STEP 3 · TEXT (3.41)

# Integrate over the rectangle and multiply by t

$$k_{34} = t \int_0^a \int_0^b \left[ \frac{E\nu}{1-\nu^2} \frac{\partial N_2}{\partial x} \frac{\partial N_2}{\partial y} + \frac{E(1-\nu)}{2(1-\nu^2)} \frac{\partial N_2}{\partial y} \frac{\partial N_2}{\partial x} \right] dy\, dx = \frac{E\, t}{8(\nu - 1)} \tag{3.41}$$

$$\frac{\partial N_2}{\partial x} \frac{\partial N_2}{\partial y} = \frac{b-y}{ab} \cdot \frac{-x}{ab} \Rightarrow \int_0^a \int_0^b \frac{-x(b-y)}{a^2 b^2}\, dy\, dx = -\frac{1}{4}$$

$$k_{34} = -\frac{E\, t}{4(1-\nu^2)} \left( \nu + \frac{1-\nu}{2} \right) = -\frac{E\, t\, (1+\nu)}{8(1-\nu^2)} = -\frac{E\, t}{8(1-\nu)}$$

**CHECK** Independent of a and b: the coupling between u and v at a node does not depend on the element's proportions. Entries on the diagonal do (next slide).

BLOCK 2 · THE WHOLE MATRIX · TEXT (3.42)

# Twelve distinct values arranged by the symmetry of the rectangle

$$[k] = \begin{bmatrix} s_1 & s_2 & s_3 & s_4 & s_5 & s_6 & s_7 & s_8 \\ s_2 & s_9 & s_8 & s_{10} & s_6 & s_{11} & s_4 & s_{12} \\ s_3 & s_8 & s_1 & s_6 & s_7 & s_4 & s_5 & s_2 \\ s_4 & s_{10} & s_6 & s_9 & s_8 & s_{12} & s_2 & s_{11} \\ s_5 & s_6 & s_7 & s_8 & s_1 & s_2 & s_3 & s_4 \\ s_6 & s_{11} & s_4 & s_{12} & s_2 & s_9 & s_8 & s_{10} \\ s_7 & s_4 & s_5 & s_2 & s_3 & s_8 & s_1 & s_6 \\ s_8 & s_{12} & s_2 & s_{11} & s_4 & s_{10} & s_6 & s_9 \end{bmatrix} \tag{3.42}$$

### Read the pattern

- Rows 1, 3, 5, 7 belong to $u_1 \ldots u_4$; rows 2, 4, 6, 8 to $v_1 \ldots v_4$.
- $k_{34} = s_6$: the entry just derived.
- Symmetric: $s_2$ at (1,2) and (2,1), and so on.
- Each s pairs the x- and y-derivatives of two shape functions through one entry of [E].

**WHY** You do not memorise this; you read it correctly from the text. The twelve formulas follow.

BLOCK 2 · THE FIRST SIX VALUES · TEXT (3.42)

# $s_1$ … $s_6$: diagonal and nearest-neighbour entries

$$s_1 = \frac{E\,t\left((1-\nu)a^2 + 2b^2\right)}{6(1-\nu^2)\,ab} \qquad s_2 = \frac{E\,t\left((1-\nu) + 2\nu\right)}{8(1-\nu^2)}$$

$$s_3 = \frac{E\,t\left((1-\nu)a^2 - 4b^2\right)}{12(1-\nu^2)\,ab} \qquad s_4 = -\frac{E\,t\left((1-\nu) - 2\nu\right)}{8(1-\nu^2)}$$

$$s_5 = -\frac{E\,t\left((1-\nu)a^2 + 2b^2\right)}{12(1-\nu^2)\,ab} \qquad s_6 = -\frac{E\,t\left((1-\nu) + 2\nu\right)}{8(1-\nu^2)}$$

**CHECK** $k_{34} = s_6 = -E\,t(1+\nu)/(8(1-\nu^2))$ — compare with step 3.

**WHY** $s_1$ depends on a/b: a long, thin element is stiffer against u-displacement along its length than across it.

BLOCK 2 · THE REMAINING SIX VALUES · TEXT (3.42)

# $s_7$ … $s_{12}$, and the row-sum check

$$s_7 = -\frac{E\,t\left((1-\nu)a^2 - b^2\right)}{6(1-\nu^2)\,ab} \qquad s_8 = \frac{E\,t\left((1-\nu) - 2\nu\right)}{8(1-\nu^2)}$$

$$s_9 = \frac{E\,t\left(2a^2 + (1-\nu)b^2\right)}{6(1-\nu^2)\,ab} \qquad s_{10} = -\frac{E\,t\left((1-\nu)b^2 - a^2\right)}{6(1-\nu^2)\,ab}$$

$$s_{11} = -\frac{E\,t\left(2a^2 + (1-\nu)b^2\right)}{12(1-\nu^2)\,ab} \qquad s_{12} = \frac{E\,t\left((1-\nu)b^2 - 4a^2\right)}{12(1-\nu^2)\,ab}$$

**CHECK** Rigid translation in x (all $u_i = 1$, all $v_i = 0$) gives zero force, so the u-entries of every row sum to zero: $s_1 + s_3 + s_5 + s_7 = 0$.

**WATCH OUT** [k] has three zero-energy modes (two translations, one rotation): singular until the supports prevent them.

# Software adds $[B]^T[E][B]$ at four Gauss points

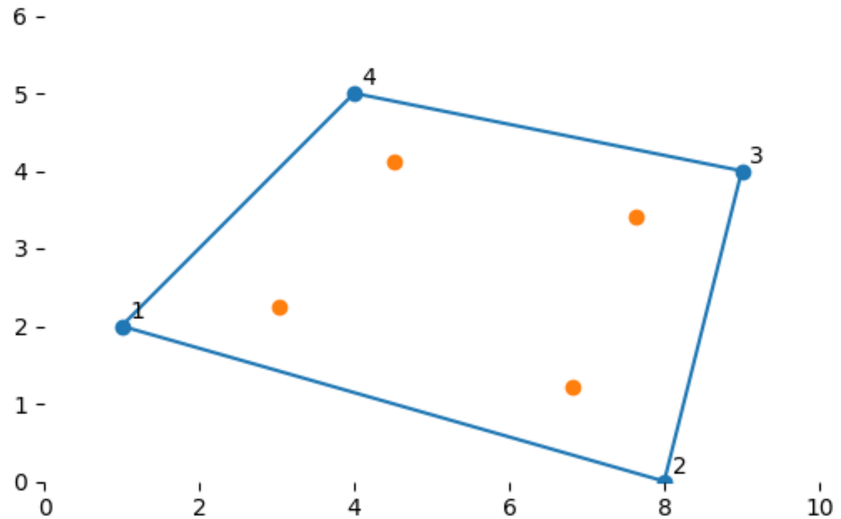



Image: PyMAPDL documentation, “Build a finite element model” (how a finite element works), © ANSYS, Inc., MIT licence.

### What the program does

- Evaluate [B] at $\xi, \eta = \pm 1/\sqrt{3}$ (the orange points).
- Form $t\,[B]^T[E][B] \times \det J$ at each point; weights are 1.
- Add the four: that is [k], 8 × 8.

**CHECK** For our rectangle the four-point sum equals the closed form (3.42) to machine precision: we checked it for the cantilever element.

**WHY** For a distorted element like this one there is no closed form, so numerical integration in $\xi, \eta$ is the only route. That is why software works on the square.

ACTIVITY 2 · PAIRS · CALCULATION

# Evaluate $s_1$, $s_2$, $s_3$ for the cantilever element and check against the text

8 MIN

**Given**

- a = 2 m, b = 1 m, t = 0.01 m, E = 200 × $10^6$ N $m^{-2}$, v = 0.33 (so 1 − $v^2$ = 0.8911).
- $s_1$, $s_2$, $s_3$ from the formulas on the previous slides.

**Find**

- E t/(1 − $v^2$), then $s_1$, $s_2$, $s_3$ in N $m^{-1}$.
- Compare with row 1 of the text's (3.44): 0.875, 0.373, −0.123 (× $10^6$).
- If time: $s_5$ and $s_7$, and check $s_1 + s_3 + s_5 + s_7 = 0$.

$$s_1 = \frac{E\,t\,\left((1-\nu)a^2 + 2b^2\right)}{6(1-\nu^2)\,ab} \qquad s_2 = \frac{E\,t\,\left((1-\nu) + 2\nu\right)}{8(1-\nu^2)}$$

$$s_3 = \frac{E\,t\,\left((1-\nu)a^2 - 4b^2\right)}{12(1-\nu^2)\,ab} \qquad s_4 = -\frac{E\,t\,\left((1-\nu) - 2\nu\right)}{8(1-\nu^2)}$$

$$s_5 = -\frac{E\,t\,\left((1-\nu)a^2 + 2b^2\right)}{12(1-\nu^2)\,ab} \qquad s_6 = -\frac{E\,t\,\left((1-\nu) + 2\nu\right)}{8(1-\nu^2)}$$

**CHECK** Units: N $m^{-2}$ × m = N $m^{-1}$ (force per unit displacement).

8 minutes. Reveal follows.

ACTIVITY 2 · SOLUTION · TEXT (3.44)

# The element stiffness matrix of the cantilever, $\times 10^6$ N m$^{-1}$

$$\frac{Et}{1-\nu^2} = \frac{200\times 10^6(0.01)}{0.8911} = 2.2444\times 10^6$$

$$s_1 = 2.2444\times 10^6\,\frac{0.67(2)^2+2(1)^2}{6(2)(1)} = 2.2444\times 10^6\,\frac{4.68}{12} = 0.875\times 10^6$$

$$s_2 = 2.2444\times 10^6\,\frac{0.67+0.66}{8} = 0.373\times 10^6, \qquad s_3 = 2.2444\times 10^6\,\frac{2.68-4}{24} = -0.123\times 10^6$$

$$[k] = \begin{bmatrix} 0.875 & 0.373 & -0.123 & -0.003 & -0.438 & -0.373 & -0.314 & 0.003 \\ 0.373 & 1.622 & 0.003 & 0.623 & -0.373 & -0.811 & -0.003 & -1.434 \\ -0.123 & 0.003 & 0.875 & -0.373 & -0.314 & -0.003 & -0.438 & 0.373 \\ -0.003 & 0.623 & -0.373 & 1.622 & 0.003 & -1.434 & 0.373 & -0.811 \\ -0.438 & -0.373 & -0.314 & 0.003 & 0.875 & 0.373 & -0.123 & -0.003 \\ -0.373 & -0.811 & -0.003 & -1.434 & 0.373 & 1.622 & 0.003 & 0.623 \\ -0.314 & -0.003 & -0.438 & 0.373 & -0.123 & 0.003 & 0.875 & -0.373 \\ 0.003 & -1.434 & 0.373 & -0.811 & -0.003 & 0.623 & -0.373 & 1.622 \end{bmatrix} \times 10^6\ \mathrm{N\,m^{-1}} \quad (3.44)$$

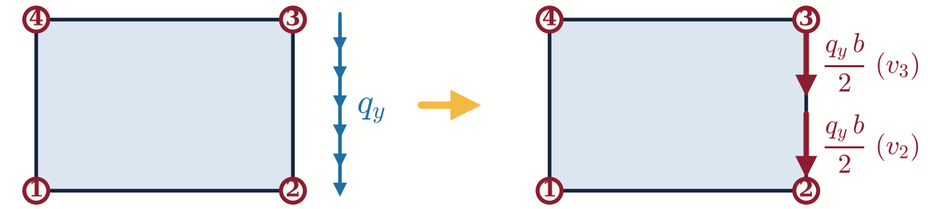
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BLOCK 3 · 1:05 – 1:25

# Load vectors

A distributed traction on an edge becomes two nodal forces; a body force becomes four. Text (3.36)–(3.37), (3.45)–(3.47).

# A uniform traction on edge 2–3 loads $v_2$ and $v_3$ equally




$$\{b\} = \int_0^b [N]^T\big|_{x=a} \begin{Bmatrix} 0 \\ q_y/t \end{Bmatrix} t\,dy = \frac{q_y\,b}{2}\{0\ \ 0\ \ 0\ \ 1\ \ 0\ \ 1\ \ 0\ \ 0\}^T$$

**WHY** $Q_y$ = −5000 N m⁻¹ (downward), b = 1 m: −2500 N at $v_2$ (dof 4) and $v_3$ (dof 6) of elements 2 and 4. Only elements with an edge on the loaded boundary get a {b}.

$$\{r^2\} = \{r^4\} = \{0\ \ 0\ \ 0\ \ \frac{Q_y b}{2}\ \ 0\ \ \frac{Q_y b}{2}\ \ 0\ \ 0\}^T = \{0\ \ 0\ \ 0\ \ -2500\ \ 0\ \ -2500\ \ 0\ \ 0\}^T\ \text{N} \quad (3.45)$$

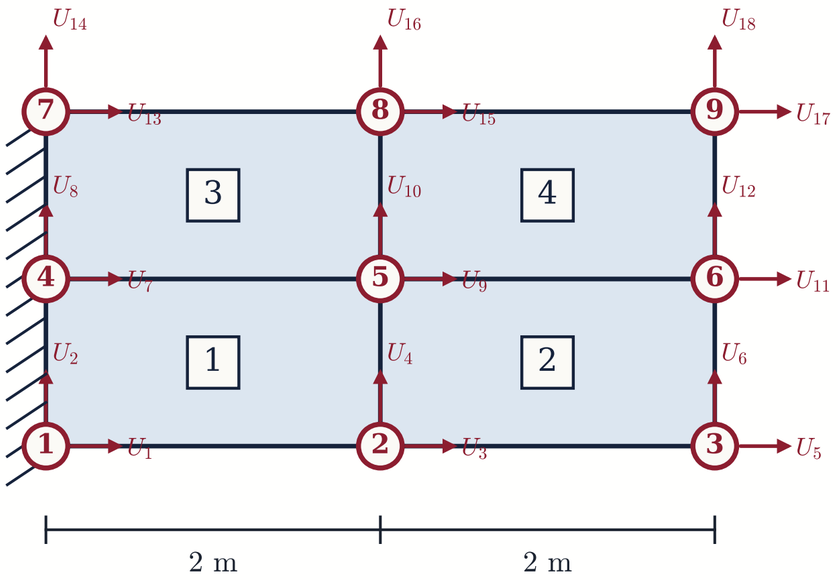
$$\{q\} = \int_x\int_y [N]^T \begin{Bmatrix} f_x \\ f_y \end{Bmatrix} t\,dy\,dx \xrightarrow{\text{self-weight } f_y=-\rho g} -\frac{\rho g\,t\,ab}{4}\{0\ \ 1\ \ 0\ \ 1\ \ 0\ \ 1\ \ 0\ \ 1\}^T$$

Body force, e.g. self-weight: a quarter of the element weight at each node, in the v-dofs.

ACTIVITY 3 · PAIRS · BOOKKEEPING

# Assemble the global load vector of the cantilever

6 MIN




**Do**

- Which elements carry the edge traction? Which of their local dofs?
- Map those local dofs to global dofs with the node numbering (node n → $U_{2n-1}$, $U_{2n}$).
- Write the 18 × 1 global load vector {R}. Which entries are unknown reactions?

6 minutes. Reveal follows, then the break.

ACTIVITY 3 · SOLUTION · TEXT (3.47)

# Three loaded dofs, six reactions, nine zeros

$$\{R\} = \{B_1\ B_2\ 0\ 0\ 0\ -2500\ B_7\ B_8\ 0\ 0\ 0\ -5000\ B_{13}\ B_{14}\ 0\ 0\ 0\ -2500\}^T\ \text{N} \quad (3.47)$$

| Element | Local loaded dofs | Global nodes | Global dofs | Load (N) |
|---|---|---|---|---|
| 2 | $v_2$, $v_3$ (local 4, 6) | 3, 6 | $U_6$, $U_{12}$ | −2500 each |
| 4 | $v_2$, $v_3$ (local 4, 6) | 6, 9 | $U_{12}$, $U_{18}$ | −2500 each |

**WHY** Node 6 is shared by both loaded elements, so $U_{12}$ receives −5000: the full 5 kN m$^{-1}$ × 1 m of edge around it.

1:25 – 1:35

# Break

Back in 10 minutes. Second half: assembly with two dofs per node, supports, the solution and its checks.

BLOCK 4 · 1:35 – 2:15

4

# Assembly and supports

Scatter-add 8 × 8 blocks with the dof map, then delete the six fixed dofs. Text (3.46)–(3.48).

BLOCK 4 · THE DOF MAP

# Connectivity in node numbers becomes connectivity in dof numbers

$$\text{node } n \;\rightarrow\; \text{dofs } 2n-1\ (u) \;\text{ and }\; 2n\ (v)$$

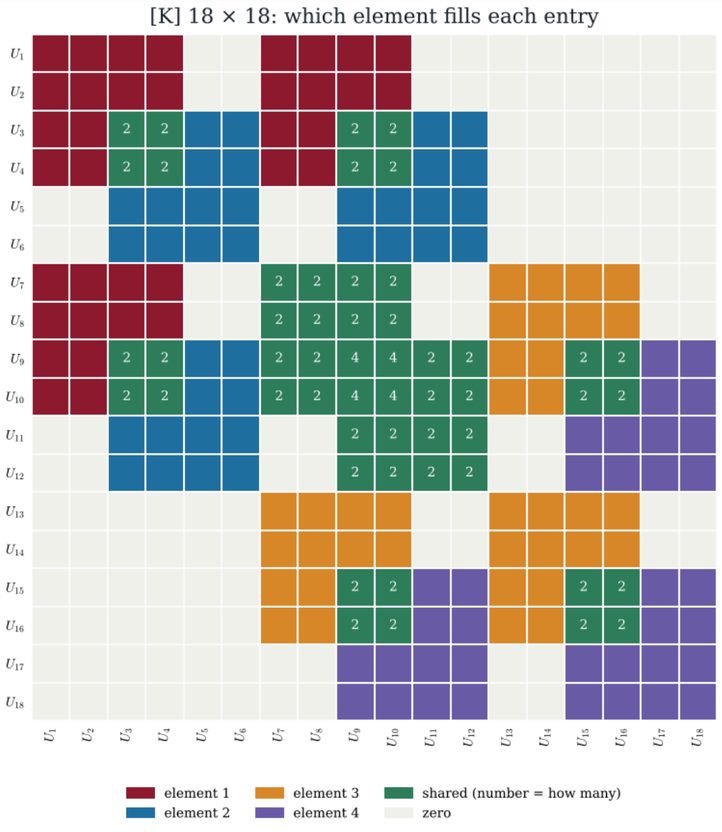
$$\text{element 1: nodes } [1\ \ 2\ \ 5\ \ 4] \;\rightarrow\; \text{dofs } [1\,2\ \ \ 3\,4\ \ \ 9\,10\ \ \ 7\,8]$$

| Element | Nodes (1)(2)(3)(4) | Dofs, in local order 1…8 |
|---|---|---|
| 1 | 1, 2, 5, 4 | 1 2 · 3 4 · 9 10 · 7 8 |
| 2 | 2, 3, 6, 5 | 3 4 · 5 6 · 11 12 · 9 10 |
| 3 | 4, 5, 8, 7 | 7 8 · 9 10 · 15 16 · 13 14 |
| 4 | 5, 6, 9, 8 | 9 10 · 11 12 · 17 18 · 15 16 |

$$K_{IJ} \;+\!=\; k^e_{ij}, \qquad I = \text{dof}(e,i),\ J = \text{dof}(e,j), \qquad i,j = 1\ldots 8$$

**WHY** MATLAB: K([1 2 3 4 9 10 7 8],[1 2 3 4 9 10 7 8]) += k — the text's §3.4.1 lines.

# Each entry is filled by the elements that share both its dofs




## Read the map

- Diagonal blocks of node 5 ($U_9$, $U_{10}$) are shared by all four elements: the number 4.
- Edge nodes are shared by two elements; corner nodes by one.
- Zero where the two dofs belong to nodes that no element has in common: node 1 and node 9 never meet.

**WHY** Bandwidth: the non-zeros stay close to the diagonal because neighbouring nodes have nearby numbers. Numbering across the short side first keeps the band narrow — software renumbers for you.

ACTIVITY 4 · PAIRS · ASSEMBLY

# Place two entries of element 2 into [K], then find the fixed dofs

8 MIN

**Do**

- Element 2 has nodes 2, 3, 6, 5. Write its dof list in local order 1…8.
- Its local $k_{11} = 0.875 \times 10^6$ and local $k_{34} = -0.373 \times 10^6$. To which global entries $K_{IJ}$ do they go?
- Which global diagonal entries receive $s_1 = 0.875 \times 10^6$ from more than one element? Give $K_{9,9}$.
- List the six fixed dofs and the size of the reduced system.

$$[k] = \begin{bmatrix} 0.875 & 0.373 & -0.123 & -0.003 & -0.438 & -0.373 & -0.314 & 0.003 \\ 0.373 & 1.622 & 0.003 & 0.623 & -0.373 & -0.811 & -0.003 & -1.434 \\ -0.123 & 0.003 & 0.875 & -0.373 & -0.314 & -0.003 & -0.438 & 0.373 \\ -0.003 & 0.623 & -0.373 & 1.622 & 0.003 & -1.434 & 0.373 & -0.811 \\ -0.438 & -0.373 & -0.314 & 0.003 & 0.875 & 0.373 & -0.123 & -0.003 \\ -0.373 & -0.811 & -0.003 & -1.434 & 0.373 & 1.622 & 0.003 & 0.623 \\ -0.314 & -0.003 & -0.438 & 0.373 & -0.123 & 0.003 & 0.875 & -0.373 \\ 0.003 & -1.434 & 0.373 & -0.811 & -0.003 & 0.623 & -0.373 & 1.622 \end{bmatrix} \times 10^6 \ \mathrm{N\,m^{-1}}$$

8 minutes. Reveal follows.

ACTIVITY 4 · SOLUTION · TEXT (3.48)

# Six fixed dofs; twelve equations remain

$$U_1 = U_2 = U_7 = U_8 = U_{13} = U_{14} = 0$$

$$\Rightarrow \text{ delete 6 rows and columns: a } 12 \times 12 \text{ system, text } (3.48)$$

| Local entry of element 2 | Global entry | Value (× $10^6$) |
|---|---|---|
| $k_{11}$ ($u_2$–$u_2$) | $K_{3,3}$ | +0.875 (plus element 1's $k_{33}$ = 0.875 → 1.751) |
| $k_{34}$ ($u_2$–$v_2$ coupling) | $K_{5,6}$ | −0.373 |
| $k_{77}$ ($u_4$–$u_4$, node 5) | $K_{9,9}$ | +0.875 (× 4 elements → 3.500) |

**CHECK** $K_{3,3}$ = 1.751 and $K_{9,9}$ = 3.500 are visible in the text's (3.46) and (3.48): a good place to check your dof map.

BLOCK 4 · THE SOLUTION · TEXT (3.49)

# Nodal displacements of the 2 × 2 mesh, in metres

| Node | x-dof | u (m) | y-dof | v (m) |
|---|---|---|---|---|
| 1 (fixed) | $U_1$ | 0 | $U_2$ | 0 |
| 2 | $U_3$ | −0.031 | $U_4$ | −0.046 |
| 3 (free end, bottom) | $U_5$ | −0.042 | $U_6$ | −0.131 |
| 4 (fixed) | $U_7$ | 0 | $U_8$ | 0 |
| 5 (centre) | $U_9$ | 0.000 | $U_{10}$ | −0.043 |
| 6 (free end, mid) | $U_{11}$ | 0.000 | $U_{12}$ | −0.130 |
| 7 (fixed) | $U_{13}$ | 0 | $U_{14}$ | 0 |
| 8 | $U_{15}$ | 0.031 | $U_{16}$ | −0.046 |
| 9 (free end, top) | $U_{17}$ | 0.042 | $U_{18}$ | −0.131 |

**WHY** Bending signature: the bottom fibre shortens ($u_3$, $u_5 < 0$), the top lengthens ($u_{15}$, $u_{17} > 0$), and mid-height u = 0.

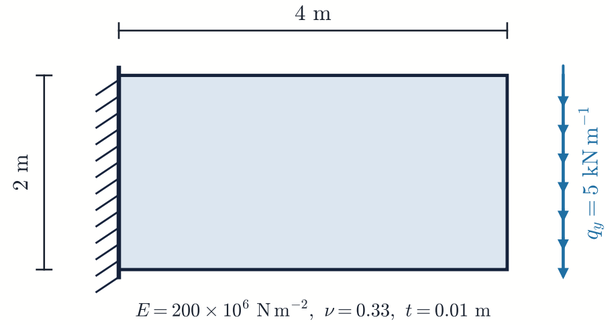
ACTIVITY 5 · PAIRS · INTERPRETATION

# Read the solution like a structural engineer

6 MIN

**From the table**

- Why is $U_5 = -U_{17}$ and $U_6 = U_{18}$ exactly?
- Why is $U_9 = U_{11} = 0$?
- The free end moves down 0.131 m. Estimate the tip deflection of the same cantilever with elementary beam theory, $\delta = PL^3/(3EI)$, with P = 10 kN, L = 4 m, $I = t\,h^3/12$. Compare.




6 minutes. Reveal follows.

ACTIVITY 5 · SOLUTION

# Antisymmetric bending, a neutral axis, and a mesh that is too stiff

$$\delta = \frac{PL^3}{3EI} + \frac{PL}{\kappa G A}, \qquad P = q_y H = 10\ \text{kN},\ I = \frac{tH^3}{12},\ A = tH,\ \kappa = \tfrac{5}{6},\ G = \frac{E}{2(1+\nu)}$$

Timoshenko beam: bending + shear

$$\delta = \frac{10^4 (4)^3}{3(200\times 10^6)(6.667\times 10^{-3})} + \frac{10^4 (4)}{\frac{5}{6}(75.19\times 10^6)(0.02)} = 0.160 + 0.032 = 0.192\ \text{m}$$

**WATCH OUT** Shear adds 0.032 m to the 0.160 m of bending: 17% for a beam this deep (L/h = 2). The 2 × 2 FE answer, 0.131 m, is 32% below the beam estimate.

**CHECK** Both symmetries are exact properties of the discrete system, not approximations: the mesh, the loads and the supports are symmetric about y = 1 m.

BLOCK 5 · 2:15 – 2:40

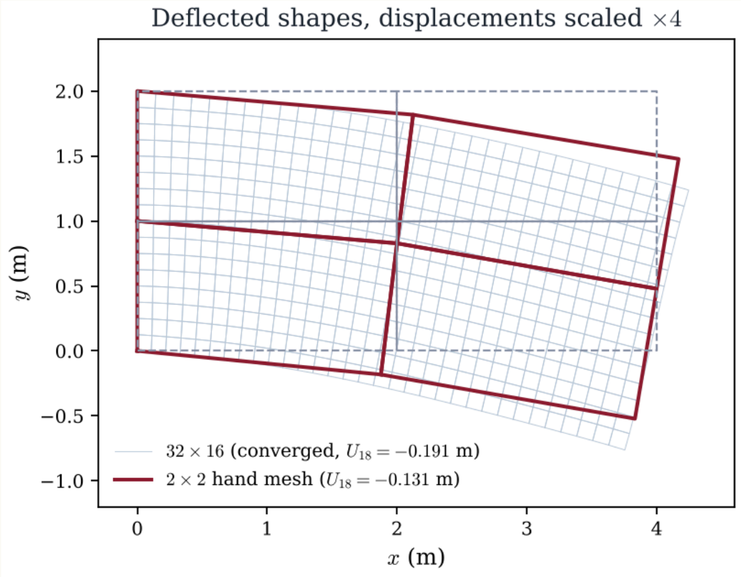
5

# Is the answer right? Refinement and a hand check

Refine the mesh until the deflection stops moving, and compare with beam theory. Text Table 3.1, Fig. 3.7.

BLOCK 5 · DEFLECTED SHAPES

# The hand mesh deflects two thirds as much as the converged mesh



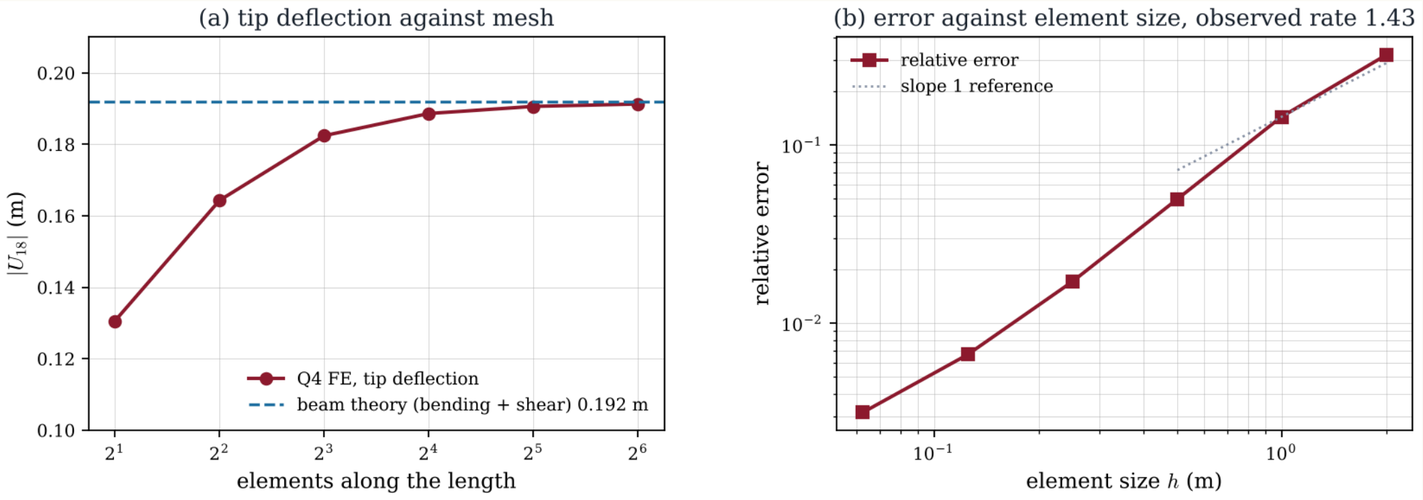

### What the picture shows

- Both meshes bend; the coarse one bends less.
- The bilinear element cannot curve: each element’s edges stay straight, so bending is represented by a kink pattern.
- Straight edges force spurious shear strain under bending — the element resists with shear stiffness it should not have: **shear locking**.

**WHY** More elements → smaller kinks → less spurious shear. The 8-node element (Plane Stress Part 3) curves instead.

BLOCK 5 · MESH CONVERGENCE · TEXT TABLE 3.1

# Refine until the tip deflection stops moving; the limit agrees with beam theory

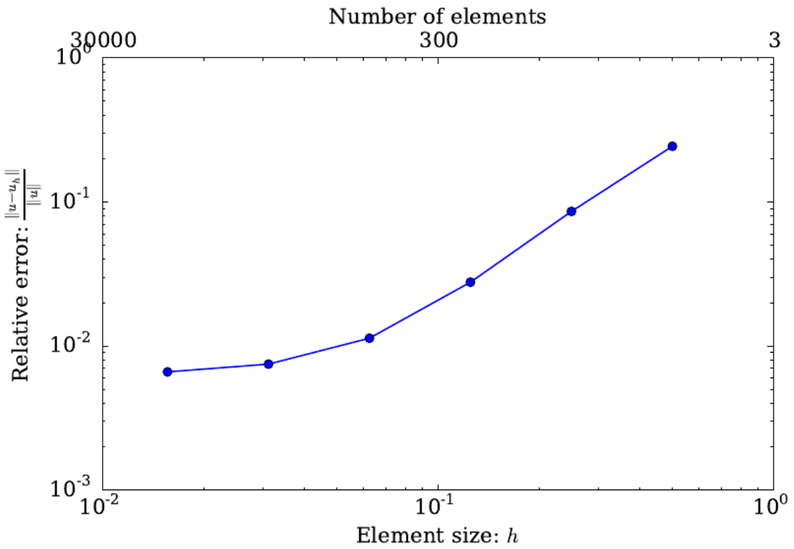


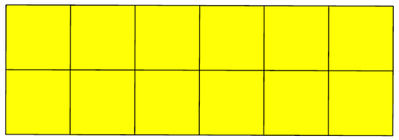

| Mesh | $U_6$ (m) | $U_{18}$ (m) |
|---|---|---|
| Q4 2 × 2 | −0.131 | −0.131 |
| Q4 16 × 16 | −0.1893 | −0.1893 |
| Q4 80 × 80 | −0.192 | −0.192 |
| Software (text) | −0.192 | −0.192 |
| Beam theory | — | −0.192 |

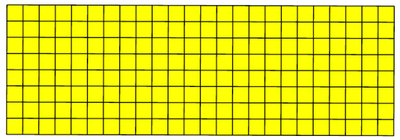
$$\frac{|U_{18}|_{2\times2}-\delta}{\delta}=\frac{0.131-0.192}{0.192}=-32\% \qquad \frac{0.1893-0.192}{0.192}=-1.4\%\ (16\times16)$$

BLOCK 5 · THE SAME EXPERIMENT IN ANOTHER CODE

# Error falls with element size, then levels off near 0.7%





**CHECK** Coarse meshes: each halving of h cuts the error about three times. Same lesson as our Table 3.1.

**WHY** Fine meshes: the error stops near 0.7%. The FE end load is spread uniformly; the exact elasticity solution assumes a parabolic end shear. That is a modelling difference, not a mesh error — as our converged 0.191 against Timoshenko's 0.192 m.

Images: SolidsPy (J. Gómez and N. Guarín-Zapata), "Convergence analysis for a cantilever beam", MIT licence, github.com/nicoguaro/SolidsPy.

ACTIVITY 6 · PAIRS · CALCULATION

# Quantify the coarse-mesh error and decide on a mesh

8 MIN

**Do**

- Percentage error of the 2 × 2 and the 16 × 16 tip deflections relative to the converged 0.192 m.
- If the error roughly halves when the element size halves, how many elements along the length are needed for 1%?
- A project team reports a shear-wall deflection from a single 4 × 6 mesh. What do you ask them to add?

$$\frac{|U_{18}|_{2\times2} - \delta}{\delta} = \frac{0.131 - 0.192}{0.192} = -32\% \qquad \frac{0.1893 - 0.192}{0.192} = -1.4\% \ (16 \times 16)$$

**CHECK** Use the table and the plot; no new theory needed.

8 minutes. Reveal follows.

ACTIVITY 6 · SOLUTION

# Report the error, refine until it is below the tolerance, and check one number by hand

### Errors

- 2 × 2: (0.131 − 0.192)/0.192 = −32%
- 16 × 16: (0.1893 − 0.192)/0.192 = −1.4%
- 32 × 16: (0.191 − 0.192)/0.192 = −0.7%

### Mesh for 1%

About 32 elements along the length (element size 0.125 m). Finer than that costs time and buys nothing a designer can use.

### The project rule

Three meshes, one table, one independent check (beam theory, equilibrium of reactions, or a textbook solution). Then — and only then — a conclusion.

**CHECK** Reactions check: the six reactions at the fixed edge must sum to +10 kN upward and give the moment 10 × 4 = 40 kN m. Software prints them; use them.

QR EXPLORER · PAIRS, ONE PHONE

# Your hand mesh, then better meshes, on your phone

8 MIN

### Scan, then do

1. 4-node, 2 × 2: does the Explorer give the text's −0.131 m? How far is it from beam theory?
2. Same eight elements, two layouts: 4 × 2 and 8 × 1, then 2 × 4. Which is stiffest, and why?
3. Find the coarsest mesh within 2% of −0.192 m with 4-node and with 8-node elements. Compare their dofs.

**Cantilever Explorer**

`razinutm.github.io/My_Lectures/SEAA4243-FEM/labs/cant.html`

8 minutes, optional in class: if the block is running late, set it as homework. The mesh study button builds the table of Activity 6 for you.

BLOCK 6 · 2:40 – 3:00

6

# The MATLAB script and what comes next

Four changes turn the heat-transfer script into the plane-stress script. Then the exit ticket.

# Four changes turn the heat-transfer script into the plane-stress script

```
% ---- changes from the heat script ----
E = 200e6; nu = 0.33; t = 0.01; qy = 5000;
Nx = 2; Ny = 2; ndof = 18; % 2 dofs per node
a = Lx/Nx; b = Ly/Ny;

% 12 closed-form values, text (3.42)
s1 = (E*t*((1-nu)*a^2 + 2*b^2))/(6*(1-nu^2)*a*b);
s2 = (E*t*((1-nu) + 2*nu))/(8*(1-nu^2));
s3 = (E*t*((1-nu)*a^2 - 4*b^2))/(12*(1-nu^2)*a*b);
% ... s4 to s12 likewise ...
k = [s1 s2 s3 s4 s5 s6 s7 s8; ...]; % 8 x 8, (3.42)
```

```
% assembly with the dof map (node n → 2n-1, 2n)
K = zeros(ndof); F = zeros(ndof,1);
d1 = [1 2 3 4 9 10 7 8]; K(d1,d1) = K(d1,d1) + k;
d2 = [3 4 5 6 11 12 9 10]; K(d2,d2) = K(d2,d2) + k;
d3 = [7 8 9 10 15 16 13 14]; K(d3,d3) = K(d3,d3) + k;
d4 = [9 10 11 12 17 18 15 16]; K(d4,d4) = K(d4,d4) + k;

% edge traction → nodal loads, text (3.47)
F([6 12 18]) = [-2500 -5000 -2500];
% fixed edge, then solve
dof_k = [1 2 7 8 13 14];
dof_u = setdiff(1:ndof, dof_k);
U = zeros(ndof,1);
U(dof_u) = K(dof_u,dof_u) \ F(dof_u);
```

Run it: U(dof_u) reproduces (3.49). Text §3.4.1 is the full listing; §3.4.2 is the 8-node version with 16 × 16 blocks.

ACTIVITY 7 · INDIVIDUAL · EXIT TICKET

# Three questions before you leave

5 MIN

$$[k] = t \int_x \int_y [B]^T [E][B]\, dy\, dx \qquad (8 \times 3)(3 \times 3)(3 \times 8) = 8 \times 8$$

**Questions**

- What are the sizes of [B], [E] and [k] for the 4-node plane-stress element, and what does t multiply?
- An edge of length b carries a uniform traction q. What does each of its two nodes receive?
- Your 2 × 2 model gives 0.131 m and the converged answer is 0.192 m. Is the coarse model too stiff or too flexible, and why?

Five minutes. Name and matric number; for feedback, not marks.

WRAP-UP · WHAT TO TAKE AWAY AND WHAT TO PREPARE

# Next session: stresses from displacements, the 8-node element, and the project briefing

### Today in four lines

- [B] = [∂]ᵀ[N] (3 × 8); [k] = t∫[B]ᵀ[E][B] dA (8 × 8), symmetric, three zero-energy modes.
- Twelve closed-form values $s_1 \ldots s_{12}$; for the cantilever element $k_{11} = 0.875 \times 10^6$ N m$^{-1}$.
- Edge traction → qb/2 at each edge node (dofs v); assembly by the dof map; six fixed dofs → 12 × 12.
- 2 × 2 mesh: tip deflection 0.131 m, 32% too stiff; converged 0.192 m = beam theory with shear.

### Before Plane Stress Part 3

- Read text §3.4 (pages 97–107) and Table 3.2 (8-node results).
- Compute [E][B] at the Heat Transfer Part 1 point with the Activity 1 [B]: Plane Stress Part 3 starts there.
- Form project groups; bring one candidate structure (shear wall, deep beam, gusset) for the briefing.

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